COUNTEREXAMPLES AND ENUMERATION FOR CONTRACTIBLE AND DISMANTLABLE GRAPHS

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ABSTRACT. Contractible transformations of graphs preserve the homotopy type of their clique complexes. Restricting to deletions, and imposing this restriction recursively on their neighbourhoods, gives strong I-contractibility. We construct 13-vertex graphs that are acyclic over $\mathbb{F}_2$ but have first integral homology $\mathbb{Z}/3$, and 14-vertex I-contractible graphs outside the strong class. Adding one admissible edge to each of the latter graphs produces a strong graph. These examples refute Conjectures 14 and 15 of Dochtermann, Espinoza, Frías-Armenta and Hernández. We give exact tests for acyclicity, strong contractibility and dismantlability, prove their correctness, and extend the cardinality table for connected graphs through order 13 and connected K_4-free graphs of order 14. The table distinguishes integral from mod-2 acyclicity and establishes minimum orders 13 and 14 for the two separations. There are exactly eight K_4-free graphs of order 14 in the non-strong I-contractible class; their minimum number of edges is 44.

1. INTRODUCTION

A contractible graph can be transformed into a single vertex by adding and deleting vertices and edges whose corresponding neighbourhoods are contractible. Requiring a reduction by deletions alone, and imposing the same requirement recursively on the neighbourhoods, gives the class of strong I-contractible graphs. What is the least order of an I-contractible graph that is not strong I-contractible? Does replacing contractibility by acyclicity over $\mathbb{F}_2$ lower this minimum?

Ivashchenko introduced contractible transformations of graphs [8, 9]. The clique complex $\mathrm{cl}(G)$ has the cliques of G as its simplices. The class $\mathfrak{I}$ of I-contractible graphs is precisely the class of graphs with contractible clique complex [4, Corollary 1.5]. Strong I-contractible graphs, denoted by $\mathfrak{I}_S$, have collapsible clique complexes [5, Theorem 1]; hence $\mathfrak{I}_S \subseteq \mathfrak{I}$.

Dochtermann, Espinoza, Frías-Armenta and Hernández [3] counted several contractible and dismantlable families through order 11. They also exhibited a 15-vertex graph D_{15} whose clique complex is a flag triangulation of the dunce hat, with $D_{15} \in \mathfrak{I} \setminus \mathfrak{I}_S$, and a one-edge extension $D_{15}^{+} \in \mathfrak{I}_S$. In our notation, their conjectures assert that these examples have minimum order:

> **Conjecture 14 of** [3, p. 8]**.** The graph D_{15} is smallest by number of vertices among graphs with trivial reduced $\mathbb{F}_2$-homology that are not strong I-contractible. In particular, it is smallest among the graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$.

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> **Conjecture 15 of** [3, p. 9]**.** The graph D_{15}^{+} is smallest by number of vertices among graphs $G \in \mathfrak{I}_S$ having an edge uv for which the common neighbourhood $N_G(u, v)$ belongs to $\mathfrak{I}_S$ but $G - uv \notin \mathfrak{I}_S$.

We refute both conjectures. Three graphs on 13 vertices have clique complexes homotopy equivalent to the Moore space $M(\mathbb{Z}/3, 1)$. Their reduced homology vanishes over $\mathbb{F}_2$, whereas their first integral homology is $\mathbb{Z}/3$. They therefore lie outside $\mathfrak{I}$ and refute the first assertion of Conjecture 14. A graph on 14 vertices belongs to $\mathfrak{I} \setminus \mathfrak{I}_S$ and refutes its second assertion. Its membership in $\mathfrak{I}$ follows from one admissible edge addition followed by a strong reduction. The same extension refutes Conjecture 15. Section 3 gives the constructions and their proofs.

Our second contribution is an extension of the cardinality table, together with algorithms and proofs that justify its computation. We count connected graphs up to isomorphism in the contractible and dismantlable families through order 13, and in the connected K_4-free universe at order 14. Integral and $\mathbb{F}_2$ acyclicity are recorded separately. The integral and I-contractible columns are derived from the $\mathbb{F}_2$ census and the classification of its non-strong graphs. The three torsion examples are exactly the $\mathbb{F}_2$-acyclic graphs outside $\mathfrak{I}_S$ through order 13. Consequently, the minimum order for the first assertion of Conjecture 14 is 13, while the minimum for its second assertion and for the property in Conjecture 15 is 14. At order 14 there are exactly eight K_4-free members of $\mathfrak{I} \setminus \mathfrak{I}_S$; the fewest edges among these eight is 44.

The table also records the hierarchy of k-dismantlable graphs introduced by Fieux and Jouve [6]. Throughout the listed universes, 2-Dis and 3-Dis coincide with the strong vertex I-contractible class $\mathfrak{I}_{SV}$. The first separation between 1-Dis and 2-Dis occurs at order 12, with 20 graphs.

The computation uses exact recursions for the strong classes and the dismantling levels. At each stage these recursions consider all possible first deletions. The remaining acyclic graphs are classified by explicit contractible transformations and integral homology. For K_4-free graphs, vertex neighbourhoods have clique complexes of dimension at most one and edge neighbourhoods are discrete; this gives local admissibility criteria and reduces the order-14 classification to graphs with no admissible deletion.

The 13-vertex construction also attains the minimum order for odd torsion in the first integral homology of a flag complex, by comparison with Adamaszek's classification [1]. The three 14-vertex graphs constructed below have contractible, non-collapsible clique complexes. The computer-assisted classification through order 13 shows that 14 is the minimum number of vertices of such a flag complex (Corollary 5.4). Earlier examples arise from Bing's house [2] and from flag triangulations of the dunce hat [6, 3]. Our classification of order-14 graphs is restricted to the K_4-free class; the minimum by number of vertices follows from the unrestricted census through order 13.

Organization. Section 2 fixes the graph operations and homological conventions. Section 3 proves the refutations using explicit examples. Sections 4 and 5 justify the enumeration and derive the table and the minimum orders. The appendix supplies the finite mathematical data used in the constructions.

Conventions and notation. Graphs are finite and simple. Their order is their number of vertices. Homology refers to the clique complex and is reduced unless stated otherwise. The empty graph belongs to none of the contractible or acyclic classes considered here. Cardinalities count isomorphism classes.

2. Graph reductions and homology

The graph operations below are determined by induced neighbourhoods. For a vertex v of G, we write $N_G(v)$ for the subgraph induced by its neighbours. For distinct vertices u, v, the common neighbourhood $N_G(u, v)$ is induced by the vertices adjacent to both. We write $G - v$ for a vertex deletion, $G - uv$ for an edge deletion, and $G + uv$ for the addition of a missing edge. The clique complex $\mathrm{cl}(G)$ consists of the cliques of G; a complex of this form is called *flag*. The clique complexes of $N_G(v)$ and $N_G(u, v)$ are the links of v and, when uv is an edge, of uv in $\mathrm{cl}(G)$.

2.1. Contractible and strong contractible graphs.

The two classes differ in the permitted operations and in the condition imposed on neighbourhoods.

Definition 2.1 ([3, Definition 1]). *The class $\mathfrak{I}$ of I-contractible graphs is the least class containing K_1 and closed under the following operations:*

(i) delete a vertex v when $N_G(v) \in \mathfrak{I}$;
(ii) add a new vertex whose neighbourhood is an induced subgraph of G belonging to $\mathfrak{I}$;
(iii) delete an edge uv when $N_G(u, v) \in \mathfrak{I}$;
(iv) add a missing edge uv when $N_G(u, v) \in \mathfrak{I}$.

The empty graph does not belong to $\mathfrak{I}$.

The operations in Definition 2.1 preserve the simple homotopy type of the clique complex [4, Theorem 2.8]; the graph formulation goes back to Ivashchenko and to Chen, Yau and Yeh [9, 2]. In particular, every member of $\mathfrak{I}$ has contractible clique complex. The converse holds as well [4, Corollary 1.5].

Definition 2.2 ([5, Definition 3 and Lemma 1]; [3, Definitions 4–5]). *The class $\mathfrak{I}_S$ of strong I-contractible graphs contains K_1 and is defined recursively, for graphs of order at least two, by*

$$G \in \mathfrak{I}_S \iff \begin{cases} G - v,\ N_G(v) \in \mathfrak{I}_S & \text{for some } v \in V(G), \quad \text{or} \\ G - uv,\ N_G(u, v) \in \mathfrak{I}_S & \text{for some } uv \in E(G). \end{cases}$$

The class $\mathfrak{I}_{SV}$ of strong vertex I-contractible graphs has the same base case and uses only the first alternative, with both smaller graphs required to belong to $\mathfrak{I}_{SV}$. Neither class contains the empty graph.

Each recursive argument has smaller $|V| + |E|$, so these definitions are well founded. Reversing a successful reduction gives the equivalent definitions by gluings in the cited papers. Induction on $|V| + |E|$ gives

$$\mathfrak{I}_{SV} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I}. \tag{1}$$

For the second inclusion, the remaining graph and the tested neighbourhood lie in $\mathfrak{I}$ by induction, and the inverse operation reconstructs the original graph by an allowed gluing.

An operation is *$\mathfrak{I}_S$-admissible* if its tested neighbourhood belongs to $\mathfrak{I}_S$. Unless a different class is named, admissibility will mean $\mathfrak{I}_S$-admissibility. A *strong reduction* is a sequence of admissible deletions ending at K_1. The empty-neighbourhood convention means that neither an isolated vertex nor an edge with empty common neighbourhood can be deleted. Every nonempty tree belongs to $\mathfrak{I}_{SV}$: deleting a leaf tests the neighbourhood K_1 and reduces the order. Conversely, a triangle-free member of $\mathfrak{I}$ is a tree, since its clique complex is connected, contractible and of dimension at most one.

Membership in $\mathfrak{I}_S$ requires the existence of a successful reduction. A different admissible first deletion can destroy that possibility, as the examples for Conjecture 15 will show. Thus the recursive test must consider all possible first deletions.

2.2. **Dismantling levels.** Vertex reductions also give a hierarchy in which the condition on the neighbourhood changes with the level.

Definition 2.3 ([6, Definition 1])**.** *A cone is a nonempty graph with a vertex adjacent to every other vertex. Cones form the class (-1)-Dis. For $k \geq 0$, a vertex v is k-dismantlable when $N_G(v)$ belongs to $(k-1)$-Dis. A graph is k-dismantlable, or belongs to k-Dis, if it can be reduced to K_1 by successive deletions of k-dismantlable vertices. The singleton belongs to every level, and the empty graph to none.*

At level zero, the condition says that some neighbour u of v is adjacent to all the other neighbours of v. Equivalently, $N_G[v] \subseteq N_G[u]$, where brackets denote closed neighbourhoods. The hierarchy satisfies

$$(-1)\text{-Dis} \subseteq 0\text{-Dis} \subseteq 1\text{-Dis} \subseteq \cdots \subseteq \mathfrak{I}_{SV} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I}; \tag{2}$$

see [6, Proposition 2] and [3, Lemma 19 and Theorem 23]. The union of the finite dismantling levels is $\mathfrak{I}_{SV}$ [3, Theorem 23].

2.3. **Coefficient choices.** We use homology to distinguish graphs whose clique complexes are contractible from graphs with the same homology over $\mathbb{F}_2$. Write $\widetilde{H}_i(G;R) = \widetilde{H}_i(\mathrm{cl}(G);R)$. For $R = \mathbb{Z}$ or $\mathbb{F}_2$, let $\mathcal{A}_R$ be the class of nonempty graphs for which $\widetilde{H}_i(G;R) = 0$ for every $i \geq 0$. The Euler characteristic is

$$\chi(G) = \sum_{i \geq 0} (-1)^i f_i(G),$$

where $f_i(G)$ counts the cliques with $i+1$ vertices.

For a nonempty finite complex K, the universal coefficient theorem [7, Theorem 3A.3] gives, for $i \geq 0$,

$$0 \longrightarrow \widetilde{H}_i(K;\mathbb{Z}) \otimes \mathbb{F}_2 \longrightarrow \widetilde{H}_i(K;\mathbb{F}_2) \longrightarrow \mathrm{Tor}_1^{\mathbb{Z}}(\widetilde{H}_{i-1}(K;\mathbb{Z}),\mathbb{F}_2) \longrightarrow 0. \tag{3}$$

We use the augmented convention $\widetilde{H}_{-1}(K;\mathbb{Z}) = 0$. Consequently,

$$\mathfrak{I}_S \subseteq \mathfrak{I} \subseteq \mathcal{A}_{\mathbb{Z}} \subseteq \mathcal{A}_{\mathbb{F}_2}. \tag{4}$$

For a connected finite complex, $\mathbb{F}_2$-acyclicity allows finite integral homology of odd order. In particular, both $(\mathbb{Z}/3) \otimes \mathbb{F}_2$ and $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/3,\mathbb{F}_2)$ vanish. This accounts for the different hypotheses in the two assertions of Conjecture 14. Throughout the article, unqualified acyclicity means membership in $\mathcal{A}_{\mathbb{F}_2}$.

2.4. **A disc with boundary identifications.** The first construction is most easily understood by obtaining its clique complex from a triangulated disc. The following criterion determines the homotopy type from the boundary identifications.

Proposition 2.4. *Let D be a triangulated closed disc and $q : D \to X$ a simplicial map. Suppose that q is injective on the interior of D and identifies X with the quotient of D obtained solely by identifying boundary points having the same image. If $S = q(\partial D)$ is a connected graph with exactly one cycle, then*

$$X \simeq S^1 \cup_d D^2,$$

where d *is the degree of the boundary map after contracting a maximal tree in* S*. For* $|d| = 1$ *the complex is contractible. For* $|d| = 3$ *it is homotopy equivalent to* $M(\mathbb{Z}/3, 1)$ *and has* $\widetilde{H}_1(X; \mathbb{Z}) \cong \mathbb{Z}/3$ *and zero reduced homology in all other degrees.*

Proof. The hypotheses identify X with the adjunction space $S \cup_f D^2$. Collapsing a maximal tree T in S is a homotopy equivalence, since T is a contractible subcomplex [7, Propositions 0.16–0.17]. The quotient is $(S/T) \cup_{pf} D^2$, with $S/T \cong S^1$. Maps $S^1 \to S^1$ are classified up to homotopy by their degree [7, Theorem 1.7], and homotopic attaching maps have homotopy equivalent adjunction spaces [7, Proposition 0.18]. This proves the asserted homotopy type. For $|d| = 1$ the map is homotopic to a homeomorphism, whose mapping cone is contractible. For $|d| = 3$ this is the defining adjunction space of $M(\mathbb{Z}/3, 1)$, and its cellular chain complex in positive degrees is $0 \to \mathbb{Z} \xrightarrow{d} \mathbb{Z} \to 0$. □

3. Counterexamples on thirteen and fourteen vertices

The two assertions in Conjecture 14 are refuted at different orders. Odd integral torsion supplies a graph with trivial homology over $\mathbb{F}_2$ outside $\mathfrak{I}_S$. A second construction belongs to $\mathfrak{I}$ but admits no first strong deletion. Adding one edge to that construction also gives a counterexample to Conjecture 15.

Throughout this section, G_{13} has vertex set $\{0, \dots, 12\}$ and G_{14}^a, G_{14}^b, G_{14}^c have vertex set $\{0, \dots, 13\}$. Their edges are defined by Table 2. The figures use these same labels.

3.1. **Odd torsion on thirteen vertices.**

The clique complex of G_{13} is obtained from a triangulated disc by wrapping its boundary three times around a four-cycle. This description accounts for the torsion and gives a direct proof of the first refutation.

Theorem 3.1. *Let* $G'_{13} = G_{13} + \{4, 7\}$ *and* $G''_{13} = G'_{13} + \{5, 6\}$*. The graphs* G_{13}, G'_{13}, G''_{13} *have* 40, 41 *and* 42 *edges, respectively, and their clique complexes are homotopy equivalent to* $M(\mathbb{Z}/3, 1)$*. In particular, they are* $\mathbb{F}_2$*-acyclic and none belongs to* $\mathfrak{I}$*. They refute the first assertion of Conjecture 14.*

Proof. The graph G_{13} is K_4-free and has 28 triangles. Each of its edges lies in two triangles, except for the four edges of 8–10–9–11–8, which lie in three. Cutting along that cycle gives the triangulated disc on the left of Figure 1. An explicit triangle list and an order of attachment proving that it is a disc are given in Appendix A.1. Its boundary labels, starting at the top and read clockwise, are

$$10, 9, 11, 8 \mid 10, 9, 11, 8 \mid 10, 9, 11, 8.$$

Each interior vertex occurs once. Identifying equally labelled boundary vertices and their corresponding boundary edges recovers precisely $\mathrm{cl}(G_{13})$. Proposition 2.4, with attaching degree 3, therefore gives

$$\mathrm{cl}(G_{13}) \simeq M(\mathbb{Z}/3, 1), \qquad \widetilde{H}_1(G_{13}; \mathbb{Z}) \cong \mathbb{Z}/3, \qquad \widetilde{H}_i(G_{13}; \mathbb{Z}) = 0 \quad (i \neq 1).$$

The universal coefficient theorem gives vanishing reduced homology over $\mathbb{F}_2$ in every degree.

For each of the two specified edge additions, the common neighbourhood at the time of addition is the singleton $\{12\}$. Both additions are admissible and preserve the homotopy type of the clique complex. Nonzero integral homology excludes all three graphs from $\mathfrak{I}$, hence also from $\mathfrak{I}_S$. Since they have fewer than the 15 vertices of the graph in Conjecture 14, they refute its first assertion. □

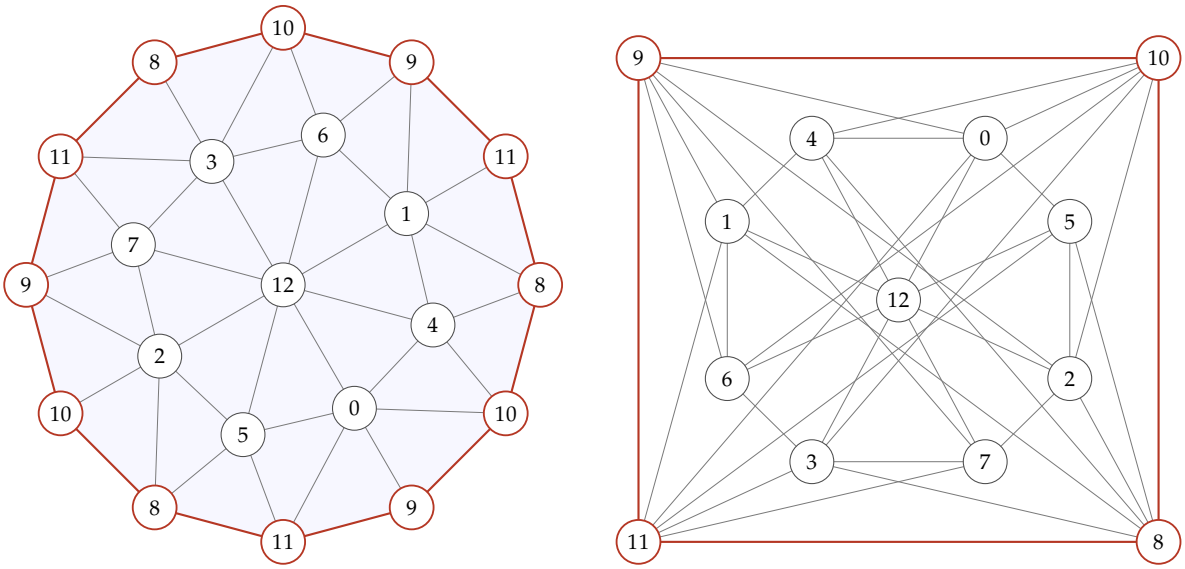


FIGURE 1. On the left, a triangulated disc whose equally labelled boundary vertices and corresponding boundary edges are identified to obtain $\mathrm{cl}(G_{13})$. The boundary traverses 10–9–11–8 three times. The 28 small triangular regions are its faces. On the right, the graph G_{13}; the four edges in three triangles are highlighted. The highlighted boundary on the left maps to the highlighted four-cycle on the right.

The construction also attains the least possible order for odd torsion in the first homology of a flag complex. This conclusion uses the earlier classification of torsion graphs, without an acyclicity assumption.

Proposition 3.2. *The minimum number of vertices of a flag complex whose first integral homology has odd torsion is* 13.

Proof. A torsion graph is called irreducible if every vertex deletion removes all torsion from its first homology. Adamaszek's classification [1, Theorem 1.1 and Section 4] gives no torsion through ten vertices, four torsion graphs on eleven vertices, each with first homology $\mathbb{Z}/2$, and 363 irreducible torsion graphs on twelve vertices, each with torsion subgroup $\mathbb{Z}/2$. The absence of torsion through ten vertices and the four torsion graphs on eleven vertices also follow from Katzman [10, Theorem 4.1 and Section 5].

If a twelve-vertex torsion graph G is not irreducible, it has a vertex v such that $G - v$ is one of the four eleven-vertex graphs. If v is isolated, then $H_1(\mathrm{cl}(G);\mathbb{Z}) = H_1(\mathrm{cl}(G-v);\mathbb{Z}) \cong \mathbb{Z}/2$. Otherwise, apply the Mayer–Vietoris sequence to

$$\mathrm{cl}(G) = \mathrm{cl}(G-v) \cup \big(v * \mathrm{cl}(N_G(v))\big).$$

It gives an exact sequence $0 \to Q \to H_1(\mathrm{cl}(G);\mathbb{Z}) \to F \to 0$, where Q is a quotient of $\mathbb{Z}/2$ and F is a subgroup of $\widetilde{H}_0(\mathrm{cl}(N_G(v));\mathbb{Z})$. The group F is free abelian, so the sequence splits. Thus $H_1(\mathrm{cl}(G);\mathbb{Z})$ has no odd torsion. Theorem 3.1 attains the lower bound. □

3.2. **A non-strong I-contractible graph on fourteen vertices.** For the second assertion of Conjecture 14, nonzero homology cannot provide the obstruction: an I-contractible graph is integrally

acyclic. Instead, we show that every possible first strong deletion is obstructed, and then establish I-contractibility by an edge addition.

Theorem 3.3. *The graph G_{14}^{a} is K_4-free, has 14 vertices and 44 edges, and belongs to $\mathfrak{I} \setminus \mathfrak{I}_S$. Its one-edge extension $G_{14}^{a} + \{0,1\}$ belongs to $\mathfrak{I}_S$, and deletion of that edge is $\mathfrak{I}_S$-admissible. Consequently, G_{14}^{a} refutes the second assertion of Conjecture 14, and $G_{14}^{a} + \{0,1\}$ refutes Conjecture 15.*

Proof. The common neighbourhood of 0 and 1 in G_{14}^{a} is the induced path

$$8\text{–}12\text{–}5\text{–}10\text{–}13.$$

Since nonempty trees belong to $\mathfrak{I}_S$ (Section 2.1), adding $\{0,1\}$ is admissible. Starting from $G_{14}^{a}+\{0,1\}$, perform the following deletions in the displayed order:

$$e(0,5),\ e(0,10),\ e(0,12),\ e(8,12),\ e(1,12),$$
$$v(5),\ v(10),\ v(3),\ v(6),\ v(2),\ v(7),\ v(1),\ v(0),$$
$$v(4),\ v(8),\ v(11),\ v(12),\ v(9).$$

Here $e(u,w)$ deletes an edge and $v(u)$ deletes a vertex; labels do not change after a deletion. Every tested neighbourhood is a nonempty tree, as the explicit lists in Table 5 show. The final graph is the singleton 13. Hence $G_{14}^{a} + \{0,1\} \in \mathfrak{I}_S$. Deleting the added edge is an I-admissible transformation, with the same common neighbourhood as before its addition, so $G_{14}^{a} \in \mathfrak{I}$.

To prove $G_{14}^{a} \notin \mathfrak{I}_S$, consider its links. Each vertex neighbourhood is a connected triangle-free graph; its first Betti number is positive in every case, as recorded in Table 4. Every edge has two or three common neighbours, and those neighbours form an independent set because G_{14}^{a} is K_4-free. Thus neither a vertex neighbourhood nor an edge neighbourhood is acyclic. None belongs to $\mathfrak{I}_S$, so no first strong deletion is possible.

The two graphs have 14 vertices, whereas both examples conjectured to be smallest in the earlier paper have 15 vertices. This proves the stated refutations. The global lower bounds are established from the complete census below. □

The order-14 classification also uses G_{14}^{b} and G_{14}^{c}, two further members of $\mathfrak{I} \setminus \mathfrak{I}_S$ that become strong I-contractible after one admissible edge addition. Table 2 defines them and Table 6 supplies their reductions.

Proposition 3.4. *The graphs G_{14}^{b} and G_{14}^{c} are K_4-free members of $\mathfrak{I} \setminus \mathfrak{I}_S$, each with 14 vertices and 45 edges. The extensions $G_{14}^{b} + \{0,3\}$ and $G_{14}^{c} + \{2,3\}$ belong to $\mathfrak{I}_S$, and the added edges are admissible.*

Proof. The common neighbourhoods for the two additions are the induced paths 7–13–9–12 and 6–13–7–10–11, respectively. In each column of Table 6, every tested neighbourhood is a nonempty tree and the last vertex is 13. This proves membership of both extensions in $\mathfrak{I}_S$ and of the original graphs in $\mathfrak{I}$. Table 4 excludes every possible first strong deletion from G_{14}^{b} and G_{14}^{c}, just as for G_{14}^{a}. The graph G_{14}^{b} has an edge with four common neighbours, whereas G_{14}^{c} has none. Together with their edge counts, this shows that G_{14}^{a}, G_{14}^{b}, G_{14}^{c} are pairwise non-isomorphic. □

The same local structure also obstructs simplicial collapses.

Proposition 3.5. *The clique complexes of G_{14}^{a}, G_{14}^{b}, G_{14}^{c} are contractible and non-collapsible.*

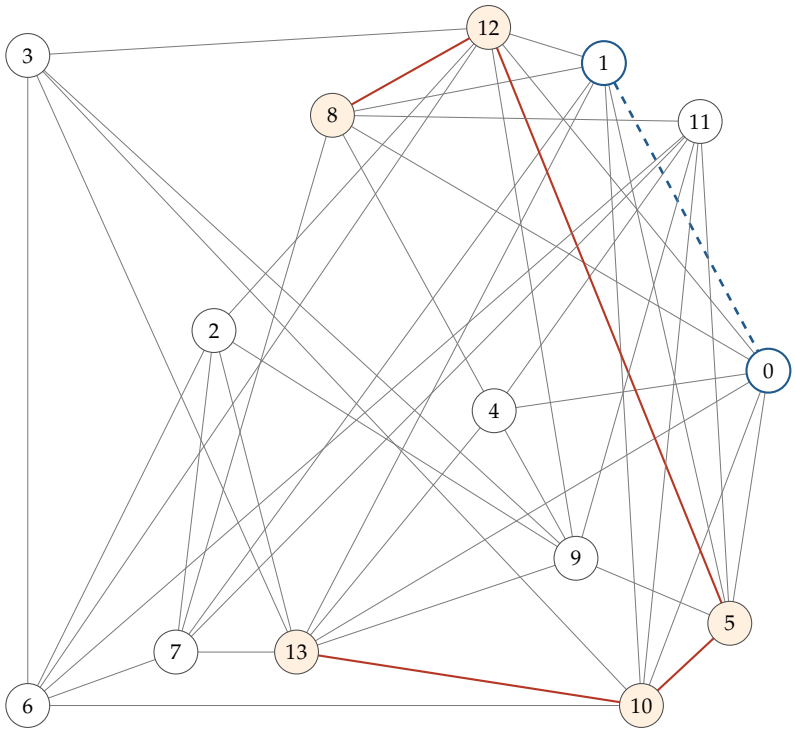


FIGURE 2. The graph G_{14}^{a} consists of the solid edges. Adding the dashed edge $\{0, 1\}$ gives a strong I-contractible graph. The common neighbourhood of its endpoints is the highlighted induced path 8–12–5–10–13. Deleting the dashed edge is admissible but leaves a graph outside $\mathfrak{I}_S$.

Proof. Their I-contractibility implies contractibility of their clique complexes. Each complex is two-dimensional. By Table 4, every edge lies in at least two triangles, and every vertex lies in a triangle. There is therefore no free face with which to start an elementary collapse. □

4. ALGORITHMS AND THE ENUMERATION OF GRAPH CLASSES

The table requires decisions about a graph and about the neighbourhoods that occur during its reductions. We first give exact procedures for these decisions. We then describe how to enumerate isomorphism classes and how to reduce the classification at order 14 to a smaller family.

4.1. **Cliques and homology.** Fix an ordering of the vertices of a nonempty graph G. Starting with each vertex, extend an increasing clique by adjoining a larger vertex adjacent to all its members, recording each clique as it is formed. Continue until no extension is possible. Every clique has a unique increasing order and a unique sequence of initial segments, so this procedure lists it exactly once. In particular, it determines all dimensions of $\mathrm{cl}(G)$, not only its triangles.

For an increasing clique $[v_0, \dots, v_i]$, use the boundary

$$\partial_i[v_0, \dots, v_i] = \sum_{j=0}^{i} (-1)^j [v_0, \dots, \widehat{v}_j, \dots, v_i].$$

Adjoin the augmentation $\partial_0 : C_0 \to \mathbb{Z}$, which sends every vertex to 1. Write $f_i = \operatorname{rank} C_i$ and let $r_i(R)$ denote the rank of ∂_i over the field R. Since consecutive boundaries compose to zero,

elementary linear algebra gives

$$\dim_R \widetilde{H}_i(G;R) = f_i - r_i(R) - r_{i+1}(R) \qquad (i \geq 0), \tag{5}$$

where $r_0(R) = 1$ and the rank above the highest dimension is zero. Thus elimination over $\mathbb{F}_2$ decides membership in $\mathcal{A}_{\mathbb{F}_2}$. The equality $\chi(G) = 1$ is a necessary preliminary condition, but is not a substitute for the homology calculation.

Integral homology also requires torsion information. If $d_{i+1,j}$ are the nonzero Smith factors of ∂_{i+1}, then

$$\widetilde{H}_i(G;\mathbb{Z}) \cong \mathbb{Z}^{\,f_i - r_i(\mathbb{Q}) - r_{i+1}(\mathbb{Q})} \oplus \bigoplus_{d_{i+1,j}>1} \mathbb{Z}/d_{i+1,j}. \tag{6}$$

Indeed, the augmented chain complex gives a short exact sequence

$$0 \longrightarrow \widetilde{H}_i(G;\mathbb{Z}) \longrightarrow C_i/\operatorname{im}\partial_{i+1} \longrightarrow \operatorname{im}\partial_i \longrightarrow 0.$$

Its last term is free abelian, so the sequence splits. The Smith factors describe the middle term, and the two adjacent rational ranks give the displayed free rank. Formula (6), in every dimension present, therefore decides integral acyclicity.

4.2. **Strong and vertex reductions.** Let $S(G)$ be a truth value. Set $S(\emptyset) = \text{false}$ and $S(K_1) = \text{true}$. For a graph with at least two vertices evaluate

$$\begin{aligned} S(G) = {} & \bigvee_{v \in V(G)} \big(S(N_G(v)) \wedge S(G - v)\big) \\ & \vee \bigvee_{uv \in E(G)} \big(S(N_G(u,v)) \wedge S(G - uv)\big). \end{aligned} \tag{7}$$

Evaluation may stop when one conjunction is true; a negative answer requires every conjunction to be false. The argument of S is always the current graph, including the edges that remain after earlier deletions.

Proposition 4.1. *The evaluation of* (7) *terminates and returns true exactly when* $G \in \mathfrak{I}_S$. *A true answer determines a reduction to* K_1 *and reductions of the neighbourhoods that justify its steps.*

Proof. Induct on $|V(G)| + |E(G)|$. Every neighbourhood and every remaining graph in (7) has smaller measure, and there are finitely many alternatives. This proves termination, starting from the two stated bases. If a conjunction is true, induction supplies the two reductions that make the corresponding first deletion valid; prepending that deletion proves $G \in \mathfrak{I}_S$. Conversely, a strong reduction of a non-singleton graph has a first vertex or edge deletion. Its neighbourhood and remainder are in $\mathfrak{I}_S$, so induction makes the corresponding conjunction true. Recording a successful choice at each occurrence gives the asserted reductions. □

For vertex reductions, all intermediate graphs are induced subgraphs of a fixed graph G. For $U \subseteq V(G)$ and $v \in U$, put $N_U(v) = U \cap V(N_G(v))$. Define truth values $W(U)$ and $D_k(U)$ to be false on the empty set and true on singletons. For nonempty U, $D_{-1}(U)$ means that $G[U]$ is a cone. On

sets of size at least two, use

$$W(U) = \bigvee_{v \in U} \big(W(N_U(v)) \wedge W(U \setminus \{v\})\big), \tag{8}$$

$$D_k(U) = \bigvee_{v \in U} \big(D_{k-1}(N_U(v)) \wedge D_k(U \setminus \{v\})\big), \qquad k \geq 0. \tag{9}$$

Proposition 4.2. *The recurrences* (8) *and* (9) *decide whether* $G[U]$ *belongs to* $\mathfrak{I}_{SV}$ *and to* k*-Dis, respectively. For a fixed largest level, evaluation in increasing order of* $|U|$ *terminates.*

Proof. Each set on the right is a proper subset of U. Induct on its cardinality, evaluating the cone test directly. A successful conjunction supplies a first admissible vertex and a reduction of the remainder. In the opposite direction, the first step of any reduction supplies one of the conjunctions. The neighbourhood condition is at the same recursive class for $\mathfrak{I}_{SV}$ and at level $k-1$ for k-Dis, exactly as in their definitions. This proves both equivalences and termination. □

At level zero there is also a simpler procedure: delete any dominated vertex until no such vertex remains. Here v is dominated by $w \neq v$ when $N_G[v] \subseteq N_G[w]$, where $N_G[v] = \{v\} \cup V(N_G(v))$. This is equivalent to $N_G(v)$ being a cone with apex w. The procedure succeeds precisely on the 0-dismantlable graphs [6, Section 4.1]. To see why the order does not matter, consider two available deletions. Unless the vertices dominate each other, each remains dominated after the other is deleted: if a dominator is removed, use its dominator and the transitivity of closed-neighbourhood inclusion. Mutual domination gives identical closed neighbourhoods, so the two remainders are isomorphic. Induction on the number of vertices now shows that an arbitrary available deletion preserves dismantlability. Reaching K_1 is itself a dismantling, proving the converse. This procedure gives the 0-Dis column of the table. No analogous order-independence holds for (7): the extensions in Section 3 admit a deletion that leaves a non-strong graph.

4.3. **Positive tests for I-contractibility.** Strong reductions also supply positive tests for the larger class $\mathfrak{I}$. Suppose

$$G = G_0, \qquad G_j = G_{j-1} + u_j v_j \quad (1 \leq j \leq t), \qquad G_t \in \mathfrak{I}_S, \tag{10}$$

and every added edge has common neighbourhood in $\mathfrak{I}_S$ at its step. Then $G \in \mathfrak{I}$. Indeed, start from $G_t \in \mathfrak{I}_S \subseteq \mathfrak{I}$ and delete the added edges in reverse order. Adding or deleting the edge between two vertices leaves their common neighbourhood unchanged, so each reverse step is an I-transformation.

For a fixed bound t, enumerate all admissible edge additions from G, one step at a time, and test each reached graph by Proposition 4.1. This finds a path of the form (10) of length at most t whenever one exists: induction on the path length shows that every possible successor is considered. The search is finite, since the vertex set is fixed and edges are only added. Its failure excludes only the specified paths, not membership in $\mathfrak{I}$. The negative decisions needed below instead use nonzero integral homology. Together these two tests will settle all graphs left outside $\mathfrak{I}_S$ in the universes of the table.

4.4. **Representatives up to isomorphism.** The generation uses canonical vertex augmentation [11, Section 4]. For each graph H of order at least two, choose a vertex orbit $O(H)$ under $\mathrm{Aut}(H)$, invariantly under isomorphism. This choice can be made from a canonical labelled representative. Begin with K_1. For every retained graph G, adjoin a vertex x adjacent to one subset from each orbit

of subsets of $V(G)$ under $\mathrm{Aut}(G)$. Retain the resulting graph H when $x \in O(H)$. Connectedness is required at the output order. For the K_4-free universe, reject augmentations containing a K_4.

Proposition 4.3. *This construction, followed by the connectedness test, generates exactly one representative of every connected graph of the requested order. The additional K_4 rejection gives exactly the connected K_4-free graphs.*

Proof. Induct on the order, before testing connectedness. For a target H, choose $x \in O(H)$. Its parent is represented by induction and the neighbourhood orbit is considered. The resulting copy of H is accepted because O is invariant under isomorphism, proving existence. For two isomorphic accepted augmentations, an isomorphism followed by an automorphism identifies the new vertices, and hence the parents and neighbourhood orbits. These were considered once, proving uniqueness. Restricting the outputs preserves both conclusions. Every induced subgraph of a K_4-free graph is K_4-free, so the additional rejection loses no predecessor of an eligible target. □

For any of the exact predicates above, sum its value over the generated representatives of a fixed order. Proposition 4.3 and the decision propositions show that this sum is the cardinality of the corresponding class, with neither missing isomorphism classes nor repeated contributions. The column for $\mathfrak{I}$ additionally requires the positive and negative adjudications described in Section 5. The inclusions in (2) and (4) allow a positive decision for one class to establish membership in every larger class of these chains; a negative decision establishes non-membership in every smaller class.

4.5. **Reduction of the order-fourteen classification.** The complete census gives the class counts. To identify the acyclic non-strong graphs behind these counts, we reduce them until no admissible move remains.

Definition 4.4. *A graph of order at least two is stuck if it is $\mathbb{F}_2$-acyclic and has no $\mathfrak{I}_S$-admissible vertex or edge. A graph of order at least two is almost stuck at u if it is $\mathbb{F}_2$-acyclic, has no $\mathfrak{I}_S$-admissible edge, and no vertex other than u is $\mathfrak{I}_S$-admissible. The distinguished vertex u need not be admissible.*

Every stuck graph lies outside $\mathfrak{I}_S$, since a reduction cannot start. Two observations explain the role of minimum order. An admissible deletion from a graph of $\mathfrak{I}$ remains in $\mathfrak{I}$. Moreover, if its remainder is in $\mathfrak{I}_S$, then the original graph is in $\mathfrak{I}_S$, by prepending that deletion to a strong reduction.

Lemma 4.5. *Let $\mathcal{H}$ be closed under vertex and edge deletion, and let n be the smallest order attained in $\mathcal{H} \cap (\mathfrak{I} \setminus \mathfrak{I}_S)$. Every graph in this class of order n reduces by admissible edge deletions to a stuck graph in the same class.*

Proof. An admissible vertex deletion would give a smaller graph in the same class, by the two observations, contradicting the choice of n. Delete admissible edges as long as possible. The order is unchanged, membership in $\mathcal{H} \cap (\mathfrak{I} \setminus \mathfrak{I}_S)$ is preserved, and the number of edges decreases. At termination neither kind of deletion is possible. The terminal graph is acyclic because it is in $\mathfrak{I}$, so it is stuck. □

Lemma 4.6. *If a stuck graph has a cut vertex u, it is the union of two induced graphs A, B, each of order at least three, that meet only at u and are almost stuck there. Consequently, if no almost stuck graph in a hereditary class has order at most m, every stuck graph in that class of order at most $2m$ is 2-connected.*

Proof. Partition the components of $G - u$ into two nonempty collections and include u in both pieces. Their clique complexes meet at one vertex, so reduced Mayer–Vietoris gives

$$\widetilde{H}_i(G;\mathbb{F}_2) \cong \widetilde{H}_i(A;\mathbb{F}_2) \oplus \widetilde{H}_i(B;\mathbb{F}_2).$$

Both pieces are acyclic. The neighbourhood of a vertex other than u, and the common neighbourhood of every edge in a piece, agree with those in G. Thus no such element is admissible. A piece of order two would have a leaf of G, which is admissible. Both orders are therefore at least three. If both exceed m, their union has order at least $2(m+1)-1 = 2m+1$, proving the last assertion. □

Lemma 4.7. *Every edge of a 2-connected stuck graph G lies in at least two triangles, and*

$$4 \le \delta(G) \le \Delta(G) \le |V(G)| - 2.$$

Proof. An edge in no triangle belongs to a cycle, by 2-connectedness. Its coefficient in every 2-boundary is zero, whereas its coefficient in that cycle is one. The cycle therefore represents nonzero homology over $\mathbb{F}_2$, contradicting acyclicity. An edge in exactly one triangle has common neighbourhood K_1 and is admissible. Hence every edge lies in at least two triangles. For $H = N_G(v)$, every vertex w of H consequently has at least two neighbours in H. If $|V(H)| \le 3$, this forces $H = K_3$, making v admissible. Thus every degree is at least four. A universal vertex would make G a cone and hence a member of $\mathfrak{I}_S$, so no degree is $|V(G)| - 1$. □

On K_4-free graphs these reductions have particularly simple tests.

Proposition 4.8. *In a K_4-free graph, a vertex is $\mathfrak{I}_S$-admissible exactly when its open neighbourhood is a nonempty tree, and an edge is $\mathfrak{I}_S$-admissible exactly when it belongs to one triangle.*

Proof. A vertex neighbourhood is triangle-free. A triangle-free graph in $\mathfrak{I}_S$ has a contractible clique complex of dimension at most one, so it is a tree. Conversely, a nonempty tree reduces by leaf deletions to K_1. The common neighbourhood of an edge is independent and belongs to $\mathfrak{I}_S$ exactly when it has one vertex. Its vertices correspond to the triangles containing the edge. □

This criterion is independent of any small-order census. In particular, to test stuckness on a K_4-free graph, check its reduced homology and exclude every tree vertex neighbourhood and every edge in one triangle. For almost stuckness allow at most one such vertex, including the possibility of none.

Finally, reverse the edge deletions of Lemma 4.5 to recover the graphs arising from a stuck graph. Starting with a finite set of stuck graphs, called *seeds*, adjoin every missing edge with common neighbourhood in $\mathfrak{I}_S$; continue from non-strong successors and discard strong ones. Identify isomorphic graphs throughout. The retained set is the *pruned edge-addition closure*.

Proposition 4.9. *This procedure terminates and retains exactly the non-strong graphs reachable from the seeds by admissible edge additions. To recover only K_4-free targets, one may also discard successors containing a K_4.*

Proof. There are finitely many graphs on the fixed vertex set and every step adds an edge. All retained graphs are reachable, by construction. If $H \in \mathfrak{I}_S$ and $H + uv$ is an admissible addition, deleting uv first and then reducing H proves $H + uv \in \mathfrak{I}_S$. A path to a non-strong target therefore never passes through a strong graph. Every predecessor on that path is expanded, so induction on path length proves completeness. Isomorphisms transport nonedges and their neighbourhoods,

and hence all possible continuations. Finally, adding edges cannot remove a K_4, so no path to a K_4-free target is lost by the additional restriction. □

5. Cardinalities and minimum orders

We count connected graphs up to isomorphism. For orders 3 through 13 the universe is unrestricted; at order 14 it consists of the connected K_4-free graphs. The exact decisions in Section 4, applied to representatives of these universes, give the directly computed columns of Table 1: connected graphs, Euler characteristic one, $\mathbb{F}_2$-acyclicity, $\mathfrak{I}_S$, $\mathfrak{I}_{SV}$, the four dismantling levels 0 through 3, and cones. The columns for $\mathfrak{I}$ and integral acyclicity are deduced below by classifying the remaining acyclic graphs.

The finite enumerations and the resulting cardinalities in this section are computer-assisted. The accompanying material records the input universes, comparisons and finite verifications, with a map from the results here to their supporting files. The mathematical procedures establish what is counted; the retained execution records supply their numerical results. The finite enumerations use the generation and isomorphism software of McKay and Piperno [12]; the versions and generation restrictions are recorded with the data.

5.1. **The first acyclic exceptions.** The directly computed acyclic and strong columns agree through order 12. At order 13 their difference is three.

Theorem 5.1. *Up to isomorphism, the only $\mathbb{F}_2$-acyclic graphs on at most 13 vertices outside $\mathfrak{I}_S$ are $G_{13}, G'_{13}, G''_{13}$. Every graph in $\mathfrak{I}$ on at most 13 vertices belongs to $\mathfrak{I}_S$, and G_{13} is the only stuck graph of order 13.*

Proof. Every $\mathbb{F}_2$-acyclic graph is connected, so the connected census contains both classes. Every strong graph is $\mathbb{F}_2$-acyclic, so equality of the two finite cardinalities through order 12 gives equality of the classes in those orders. The cases of orders one and two are immediate. At order 13 the difference is

$$5{,}619{,}022{,}190{,}286 - 5{,}619{,}022{,}190{,}283 = 3.$$

Theorem 3.1 supplies three acyclic non-strong graphs with different edge counts, hence pairwise non-isomorphic. They exhaust the difference. Their nonzero integral homology excludes them from $\mathfrak{I}$, which lies between $\mathfrak{I}_S$ and $\mathcal{A}_{\mathbb{F}_2}$. Finally, the link calculations in Table 4 show that G_{13} is stuck, whereas G'_{13} and G''_{13} each admit the edge deletion reversing the last specified addition. Every stuck graph is acyclic and non-strong, so there are no others at this order. □

5.2. **The K_4-free graphs of order fourteen.** The following classification supplies the positive exceptions in the last row of the table. Its reduction uses only the complete classification through order 13 and the local K_4-free criterion.

Theorem 5.2. *There are 51 isomorphism classes of stuck K_4-free graphs of order 14. Exactly three lie in $\mathfrak{I}$, namely $G^a_{14}, G^b_{14}, G^c_{14}$; each of the other 48 has first integral homology $\mathbb{Z}/3$ and zero higher homology. There are exactly eight K_4-free graphs of order 14 in $\mathfrak{I} \setminus \mathfrak{I}_S$, up to isomorphism. Their minimum number of edges is 44, attained by G^a_{14}.*

Proof. First, enumerate the connected K_4-free graphs of orders two through seven and apply Proposition 4.8. No graph is almost stuck. This check examines, respectively, $1, 2, 5, 17, 82, 536$ isomorphism classes; it includes graphs with no admissible vertex. If a stuck K_4-free graph of

order at most 14 had a cut vertex, Lemma 4.6 would give two almost stuck pieces with orders summing to at most 15. One would have order at most seven, a contradiction. The seeds are therefore 2-connected and satisfy $4 \leq \delta \leq \Delta \leq 12$, by Lemma 4.7.

Enumerating the K_4-free graphs of order 14 with these restrictions, and testing $\mathbb{F}_2$-homology and the local neighbourhood conditions, gives 51 pairwise non-isomorphic stuck graphs. Theorem 3.3 and Proposition 3.4 prove that $G_{14}^a, G_{14}^b, G_{14}^c$ lie in $\mathfrak{I}$. Integral homology of the other 48, computed by (6), has free rank zero in degree one, and ∂_2 has a single nonunit Smith factor, equal to 3. Thus $H_1 \cong \mathbb{Z}/3$. Their clique complexes are connected, have dimension at most two and Euler characteristic one. Hence $\widetilde{H}_0 = 0$, and H_2, which is free, has rank zero. All higher homology therefore vanishes. In particular, none of these 48 seeds lies in $\mathfrak{I}$. The list of seeds, together with its homology and the finite checks, is included in the accompanying repository.

By Theorem 5.1, a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ has at least 14 vertices. Apply Lemma 4.5 to the K_4-free class: every K_4-free member of $\mathfrak{I} \setminus \mathfrak{I}_S$ of order 14 reduces by edge deletions to a stuck graph in $\mathfrak{I}$. It therefore arises from one of $G_{14}^a, G_{14}^b, G_{14}^c$. Their pruned edge-addition closures, restricted to K_4-free graphs, have the following numbers of non-strong classes, grouped by edge count:

seed	44	45	46	47	total
G_{14}^a	1	2	1	0	4
G_{14}^b	0	1	1	1	3
G_{14}^c	0	1	0	0	1

The three sets are pairwise disjoint up to isomorphism, so their union has eight classes. Proposition 4.9 shows that no non-strong target is lost. Every retained graph belongs to $\mathfrak{I}$, since the additions are I-transformations; every possible target has now been included. This proves the count of eight.

The three contractible seeds have $44, 45, 45$ edges. Edge additions cannot decrease these numbers, so every target has at least 44 edges. The graph G_{14}^a attains the bound. □

The eight graphs have one-edge extensions in $\mathfrak{I}_S$, each obtained by an admissible addition. Thus their membership in $\mathfrak{I}$ can also be checked individually by (10); the full lists and reductions are supplied with the repository.

5.3. **Adjudication of the integral and contractible columns.** The coefficient distinction is essential to the first conjecture. Within each finite universe, the inclusions

$$\mathfrak{I}_S \subseteq \mathfrak{I} \subseteq \mathcal{A}_{\mathbb{Z}} \subseteq \mathcal{A}_{\mathbb{F}_2} \tag{11}$$

leave only the acyclic non-strong graphs to be decided. Through order 12 there are none. At order 13 all three have integral torsion, as established above.

At order 14, let G be K_4-free, $\mathbb{F}_2$-acyclic and outside $\mathfrak{I}_S$. An admissible deletion preserves its homotopy type and cannot leave a strong graph, since that would give a strong reduction of G. Delete admissible edges until either a vertex deletion is possible or no admissible element remains. In the first case the vertex deletion gives an acyclic non-strong graph of order 13. Theorem 5.1 identifies it with one of the three torsion graphs. Hence G also has nonzero integral homology. In the second case the terminal graph is one of the 51 stuck seeds. A torsion seed again gives nonzero integral homology. A contractible seed gives one of the eight graphs of Theorem 5.2. This exhausts the possibilities. In every torsion case the first integral homology is $\mathbb{Z}/3$ and the reduced homology vanishes in all other degrees.

Proposition 5.3. *In each universe of Table 1, integral acyclicity and I-contractibility coincide. Their common cardinality equals the $\mathbb{F}_2$-acyclic count through order 12, that count minus three at order 13, and that count minus 625 in the K_4-free row of order 14.*

Proof. The preceding classification partitions every acyclic non-strong graph into a member of $\mathfrak{I}$ or a graph with nonzero integral homology. Together with (11), this proves the equality of classes in the specified universes. At order 14, the acyclic and strong counts differ by

$$755{,}877{,}673{,}752 - 755{,}877{,}673{,}119 = 633.$$

Exactly eight of these graphs lie in $\mathfrak{I}$, leaving 625 with nonzero integral homology. The preceding reduction also determines their homology: $\widetilde{H}_1 \cong \mathbb{Z}/3$, with vanishing reduced homology in all other degrees. Separate integral calculations on the 625 graphs confirm these groups. Thus the integral entries are 5,619,022,190,283 at order 13 and 755,877,673,127 in the last row. All earlier entries follow from equality of the acyclic and strong classes. □

The equality just proved concerns the listed finite universes; it does not identify integral acyclicity and contractibility in general.

Table 1 combines these deductions with the direct counts. The 63 entries shared with the published table [3, Table 1] agree: these are the connected, $\mathcal{A}_{\mathbb{F}_2}$, $\mathfrak{I}$, $\mathfrak{I}_S$, $\mathfrak{I}_{SV}$, 1-Dis and 0-Dis columns at orders 3 through 11.

5.4. **Consequences of the table.** The constructions and the complete lower-order counts now determine the three minima relevant to the conjectures.

Corollary 5.4 (Computer-assisted)**.** *The minimum order of an $\mathbb{F}_2$-acyclic graph outside $\mathfrak{I}_S$ is 13. The minimum order of a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ is 14, and the same minimum holds for an integrally acyclic graph outside $\mathfrak{I}_S$. The minimum order of a strong graph having an $\mathfrak{I}_S$-admissible edge whose deletion leaves a non-strong graph is also 14. The minimum number of vertices of a contractible, non-collapsible flag simplicial complex is 14.*

Proof. Theorem 5.1 excludes the smaller orders, and Theorems 3.1 and 3.3 provide the first two examples. Proposition 5.3 excludes an integrally acyclic non-strong graph through order 13, while $G_{14}^{a} \in \mathfrak{I}$ attains order 14. Finally, an admissible edge deletion from a strong graph is an I-transformation. If its result is non-strong, that result lies in $\mathfrak{I} \setminus \mathfrak{I}_S$ and has the same order. The lower bound is therefore 14; the one-edge extension of G_{14}^{a} in Section 3 attains it.

If a contractible flag complex $K = \mathrm{cl}(G)$ has at most 13 vertices, then $G \in \mathfrak{I}$ by [4, Corollary 1.5]. Theorem 5.1 gives $G \in \mathfrak{I}_S$, and [5, Theorem 1] shows that K is collapsible. Proposition 3.5 supplies the example $\mathrm{cl}(G_{14}^{a})$ on 14 vertices. □

The dismantling columns give further separations. The classes 2-Dis and 3-Dis have the same counts as $\mathfrak{I}_{SV}$ throughout the table. The first difference between 2-Dis and 1-Dis occurs at order 12, where it has cardinality 20; at order 13 the difference is 35,757. These conclusions follow by subtracting the counts of nested classes. The 20 graphs of order 12 are catalogued in the accompanying material. The first difference between $\mathfrak{I}_S$ and $\mathfrak{I}_{SV}$, already established in [3, Theorem 11], occurs at order 11 and consists of twelve graphs.

There is also a direct interpretation of the cone column. Adjoining a universal vertex is a bijection between isomorphism classes of graphs on $n-1$ vertices and cones on n vertices. Its inverse is well

TABLE 1. Numbers of isomorphism classes of connected graphs. The two panels give twelve counts, with n repeated. The entire row $14^{\dagger}$ is restricted to K_4-free graphs. $\mathcal{A}_R$ denotes reduced acyclicity over R, and (-1)-Dis denotes the cones.

n	connected	$\chi = 1$	$\mathcal{A}_{\mathbb{F}_2}$	$\mathcal{A}_{\mathbb{Z}}$	$\mathfrak{I}$	$\mathfrak{I}_S$
3	2	2	2	2	2	2
4	6	5	5	5	5	5
5	21	16	16	16	16	16
6	112	68	68	68	68	68
7	853	404	403	403	403	403
8	11,117	3,837	3,798	3,798	3,798	3,798
9	261,080	68,458	66,167	66,167	66,167	66,167
10	11,716,571	2,571,263	2,343,148	2,343,148	2,343,148	2,343,148
11	1,006,700,565	204,131,202	166,978,401	166,978,401	166,978,401	166,978,401
12	164,059,830,476	32,482,574,568	22,666,537,431	22,666,537,431	22,666,537,431	22,666,537,431
13	50,335,907,869,219	9,859,955,384,222	5,619,022,190,286	5,619,022,190,283	5,619,022,190,283	5,619,022,190,283
$14^{\dagger}$	309,317,860,269,934	6,112,638,767,338	755,877,673,752	755,877,673,127	755,877,673,127	755,877,673,119

n	$\mathfrak{I}_{SV}$	3-Dis	2-Dis	1-Dis	0-Dis	(-1)-Dis
3	2	2	2	2	2	2
4	5	5	5	5	5	4
5	16	16	16	16	16	11
6	68	68	68	68	68	34
7	403	403	403	403	403	156
8	3,798	3,798	3,798	3,798	3,791	1,044
9	66,167	66,167	66,167	66,167	65,561	12,346
10	2,343,148	2,343,148	2,343,148	2,343,148	2,258,314	274,668
11	166,978,389	166,978,389	166,978,389	166,978,389	151,084,672	12,005,168
12	22,666,525,713	22,666,525,713	22,666,525,713	22,666,525,693	18,510,237,804	1,018,997,864
13	5,619,007,241,580	5,619,007,241,580	5,619,007,241,580	5,619,007,205,823	4,015,494,549,880	165,091,172,592
$14^{\dagger}$	755,266,302,709	755,266,302,709	755,266,302,709	755,266,302,709	81,156,275,851	20,797,002

defined because deleting any of several universal vertices gives isomorphic graphs. This explains the agreement of the unrestricted cone entries with the known numbers of unlabelled graphs [13]; the connected totals at orders 3 through 13 agree with [14]. A cone is K_4-free exactly when deletion of its apex leaves a triangle-free graph. The last cone entry is therefore 20,797,002, the number of triangle-free graphs on 13 vertices [15].

APPENDIX A. INCIDENCE DATA AND REDUCTIONS OF THE WITNESSES

The tables below specify the four graphs used in the constructions and the finite calculations in their proofs. In particular, their definitions do not depend on recovering edges from a drawing.

A.1. **A triangulated disc for the thirteen-vertex graph.** Here is an incidence description of the disc in Figure 1. Its vertices are $0, 1, \ldots, 7, 12$ and $b_0, \ldots, b_{11}$; its triangles are listed in Table 3. In the figure, b_0 is the topmost boundary vertex, and the boundary indices increase clockwise. The

TABLE 2. For each vertex i, the entries are precisely its neighbours larger than i. Thus each edge occurs once. The symbol $\varnothing$ means that there is no larger neighbour; the dash indicates that the vertex is absent.

i	G_{13}	G_{14}^{a}	G_{14}^{b}	G_{14}^{c}
0	$4,5,9,10,11,12$	$4,5,8,10,12,13$	$5,7,9,12,13$	$4,5,7,9,12,13$
1	$4,6,8,9,11,12$	$5,7,8,10,12,13$	$5,8,9,11,13$	$4,5,8,10,12,13$
2	$5,7,8,9,10,12$	$6,7,9,12,13$	$6,7,8,11,12,13$	$6,7,8,10,11,13$
3	$6,7,8,10,11,12$	$6,9,10,12,13$	$6,7,9,10,12,13$	$6,7,9,10,11,13$
4	$8,10,12$	$8,9,11,13$	$6,8,9,10,11,13$	$8,9,11,13$
5	$8,11,12$	$9,10,11,12$	$7,8,10,11,12$	$9,10,11,13$
6	$9,10,12$	$7,10,11,12$	$10,11,12$	$8,9,12,13$
7	$9,11,12$	$8,11,13$	$10,11,13$	$10,12,13$
8	$10,11$	$11,12$	$10,12,13$	$11,12$
9	$10,11$	$11,12,13$	$11,12,13$	$11,12$
10	$\varnothing$	$11,13$	$\varnothing$	$11,12$
11	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
12	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
13	—	$\varnothing$	$\varnothing$	$\varnothing$

interior vertices keep their labels. On the boundary, the map to G_{13} is

$$b_{4r} \longmapsto 10, \qquad b_{4r+1} \longmapsto 9, \qquad b_{4r+2} \longmapsto 11, \qquad b_{4r+3} \longmapsto 8 \qquad (0 \le r \le 2).$$

TABLE 3. The triangles of the disc whose boundary is attached with degree three.

j	T_j	j	T_j	j	T_j	j	T_j
1	$\{0,4,b_4\}$	8	$\{1,4,12\}$	15	$\{2,7,b_9\}$	22	$\{3,7,12\}$
2	$\{0,4,12\}$	9	$\{1,6,b_1\}$	16	$\{2,7,12\}$	23	$\{3,b_{11},b_0\}$
3	$\{0,5,b_6\}$	10	$\{1,6,12\}$	17	$\{2,b_7,b_8\}$	24	$\{3,b_{11},b_{10}\}$
4	$\{0,5,12\}$	11	$\{1,b_3,b_2\}$	18	$\{2,b_9,b_8\}$	25	$\{4,b_3,b_4\}$
5	$\{0,b_5,b_4\}$	12	$\{1,b_1,b_2\}$	19	$\{3,6,b_0\}$	26	$\{5,b_7,b_6\}$
6	$\{0,b_5,b_6\}$	13	$\{2,5,b_7\}$	20	$\{3,6,12\}$	27	$\{6,b_1,b_0\}$
7	$\{1,4,b_3\}$	14	$\{2,5,12\}$	21	$\{3,7,b_{10}\}$	28	$\{7,b_9,b_{10}\}$

To verify the disc assertion, start with T_1 and attach the remaining triangles in this order:

$$2,4,3,5,6,8,7,10,9,11,12,14,13,16,15,17,18,$$
$$20,19,22,21,23,24,25,26,27,28.$$

At each step, the intersection with the preceding union is either one boundary edge or two consecutive boundary edges, with no additional vertices. This follows directly from the triangle list. Attaching a triangle along either such boundary arc preserves a closed disc. The final boundary is $b_0, b_1, \ldots, b_{11}, b_0$.

The triangles map bijectively onto the 28 triangles of $\mathrm{cl}(G_{13})$. Their edges give exactly the adjacency data for G_{13}, which contain no K_4, so the image is the entire clique complex. Each interior vertex has a distinct label outside $\{8,9,10,11\}$. The images of interior edges are distinct, and each has an endpoint outside the four-cycle. Thus only equal-labelled boundary vertices and their

corresponding boundary edges are identified. This verifies the hypotheses of Proposition 2.4, with boundary degree three.

A.2. **The obstruction to a first deletion.** The graphs in Table 4 are K_4-free and have connected vertex neighbourhoods. The table gives their first Betti numbers $|E(N_G(v))| - |V(N_G(v))| + 1$ in vertex-label order. All are positive; every edge neighbourhood is independent with at least two vertices. Hence all these neighbourhoods have nonzero reduced homology and none belongs to $\mathfrak{I}_S$.

TABLE 4. Nonzero homology of the vertex and edge neighbourhoods. The Betti numbers of vertex neighbourhoods have the same values over every field.

Graph	First Betti numbers of vertex neighbourhoods	Edge-neighbourhood sizes
G_{13}	$1,1,1,1,1,1,1,1,2,2,2,2,1$	$2,3$
G_{14}^{a}	$1,1,1,1,1,2,1,1,1,2,2,1,2,2$	$2,3$
G_{14}^{b}	$1,1,1,1,1,2,1,2,2,2,1,1,1,3$	$2,3,4$
G_{14}^{c}	$2,1,1,1,1,1,1,2,1,2,2,2,1,2$	$2,3$

A.3. **Reductions of the one-edge extensions.** Tables 5 and 6 give the local neighbourhood at every deletion. A displayed path specifies exactly its consecutive edges. A single label denotes a one-vertex graph. At step 13 for $G_{14}^{b} + \{0,3\}$, the two displayed paths are united at their common vertices; their union is a tree. All neighbourhoods are evaluated before the corresponding deletion, and all three reductions end at the vertex 13.

TABLE 5. The complete reduction of $G_{14}^{a} + \{0,1\}$ used in Theorem 3.3.

Step	Delete	Current neighbourhood
1	$e(0,5)$	10–1–12
2	$e(0,10)$	1–13
3	$e(0,12)$	1–8
4	$e(8,12)$	1
5	$e(1,12)$	5
6	$v(5)$	1–10–11–9–12
7	$v(10)$	1–13–3–6–11
8	$v(3)$	6–12–9–13
9	$v(6)$	11–7–2–12
10	$v(2)$	7–13–9–12
11	$v(7)$	11–8–1–13
12	$v(1)$	8–0–13
13	$v(0)$	8–4–13
14	$v(4)$	8–11–9–13
15	$v(8)$	11
16	$v(11)$	9
17	$v(12)$	9
18	$v(9)$	13

TABLE 6. Reductions of the other two contractible seeds after one admissible edge addition. A dash means that the singleton has already been reached.

	$G_{14}^{b}+\{0,3\}$		$G_{14}^{c}+\{2,3\}$	
Step	Delete	Current neighbourhood	Delete	Current neighbourhood
1	$e(0,9)$	12–3–13	$e(2,7)$	10–3–13
2	$e(0,13)$	3–7	$e(2,10)$	3–11
3	$e(9,12)$	3	$e(2,13)$	3–6
4	$e(3,9)$	13	$e(6,13)$	3
5	$e(3,13)$	7	$e(3,13)$	7
6	$e(7,13)$	2	$v(7)$	3–10–12–0–13
7	$v(2)$	7–11–6–12–8–13	$v(10)$	3–11–5–1–12
8	$v(6)$	11–4–10–3–12	$v(1)$	5–13–4–8–12
9	$v(3)$	10–7–0–12	$v(4)$	8–11–9–0–13
10	$v(0)$	7–5–12	$v(0)$	12–9–5–13
11	$v(4)$	10–8–13–9–11	$v(8)$	11–2–6–12
12	$v(7)$	10–5–11	$v(2)$	6–3–11
13	$v(5)$	11–1–8–10, 8–12	$v(3)$	6–9–11
14	$v(1)$	8–13–9–11	$v(6)$	9–12
15	$v(10)$	8	$v(11)$	5–9
16	$v(11)$	9	$v(12)$	9
17	$v(9)$	13	$v(9)$	5
18	$v(12)$	8	$v(5)$	13
19	$v(8)$	13	—	—

COMPUTATIONAL MATERIAL

The accompanying reproducibility package contains the implementations, catalogues of exceptional graphs, reduction sequences, homology computations and enumeration records supporting the finite results. It provides local checks of the examples and instructions for reproducing the complete censuses. The project repository is `minimal-I-contractible-not-strong`.

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